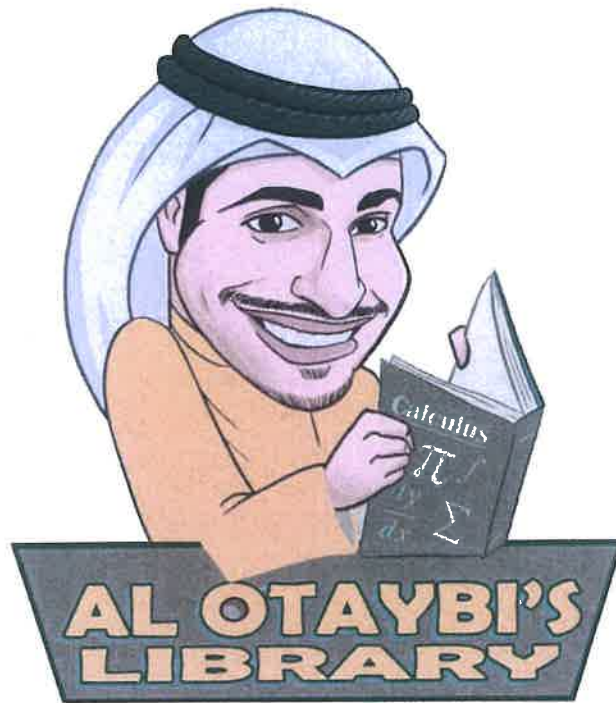


Kuwait University

2021/2022

Spring course



**Calculus B**

**First**

**Section : 4.4**

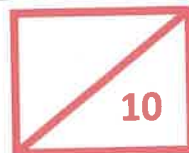
**4.4 Indeterminate Forms and l'Hospital's Rule.**

**( 2 )**

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## Quiz #1



$$\int (\sin 5x + \tan 2x) dx$$

$$\int (2 + 2x + e^{2x} - 2^{7x}) dx$$

$$\int \left( \frac{\cos x}{\sin x + 3} + \frac{3 \cos x}{\sin x + 3} - \frac{\cos x}{3 \sin x + 3} \right) dx$$

$$\int \frac{e^{2x}}{e^{2x} + 9} dx$$

$$\int \frac{e^x}{e^{2x} + 9} dx$$

$$\int \frac{e^{\sin^{-1} x}}{\sqrt{1 - x^2}} dx$$

$$\int e^x \sec(e^x) dx$$

Find  $\frac{dy}{dx}$

$$y = \cos^{-1}(\ln x) + \sec^{-1} e^x$$

Evaluate the limit

$$\lim_{x \rightarrow \infty} \ln x$$

$$\lim_{x \rightarrow \infty} \tan^{-1} x$$





## L'Hospital's Rule



قاعدة تستخدم لتحويل سؤال اللمت المعقد الى سؤال لمت يمكن حله عن طريق اشتقاق البسط المقام ( كلا على حدة )

بشرط

أن يكون ناتج التعويض المباشر

$$\frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

Definition



## L'Hospital's Rule

Definition



Suppose  $f$  and  $g$  are differentiable and  $g'(x) \neq 0$  on an open interval  $I$  that contains  $a$  (except possibly at  $a$ ). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \text{ and } \lim_{x \rightarrow a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$ ) Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

If the limit on the right side exists (or is  $\infty$  or  $-\infty$ )

# خطوات حل مسائل اللمت

## التعويض المباشر

$$0^0 \text{ أو } \infty^0 \text{ أو } 1^\infty$$

$$0 \cdot \infty$$

$$\infty - \infty$$

$$\frac{0}{0} \text{ أو } \frac{\infty}{\infty}$$

$$\text{رقم أو } \infty$$

$$1) y = \dots\dots\dots$$

$$2) \ln y = \ln \dots\dots\dots$$

أخذ اللمت للطرف اليمين (3)

النتائج =

**Final Answer**

$e$

أخذي دالة  
مقام للثانية

توحيد  
المقامات

تطبيق  
L'Hospital  
Rule

Stop !!

**Note**  
إذا عني

$e$

هي المقام

$\ln$

هي البسط



**Trick**



جزء من التعويض شكل  
اللوبيتال

يتم تطبيق اللوبيتال على هذا  
الجزء لـ

بدون عوار

راس

معرفة أنا

وين

Do Not Forget !!

$$\frac{0}{\text{عدد}} = 0, \quad \frac{\text{عدد}}{\infty} = 0, \quad \frac{\text{عدد}}{0} = \infty$$



## Text Book Question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{x^3 - 2x^2 + 1}{x^3 - 1}$$

$$\frac{1-2+1}{1-1} = \frac{0}{0}$$

L<sub>1</sub>Hop

$$\lim_{x \rightarrow 1} \frac{3x^2 - 4x}{3x^2} = \frac{3(1) - 4(1)}{3(1)} = \frac{-1}{3}$$

$$\therefore \lim_{x \rightarrow 1} \frac{x^3 - 2x^2 + 1}{x^3 - 1} = \frac{-1}{3}$$



مصيبة .. جريمة .. كارثة !!



L<sub>1</sub>HOP

$$\lim_{x \rightarrow 1} \frac{(x^3 - 1)(3x^2 - 4x) - (x^3 - 2x^2 + 1)(3x^2)}{(x^3 - 1)^2}$$

حل خاطئ !



L<sub>1</sub>Hospital Rule

ليست اشتقاق البسط والمقام بقاعدة القسمة  
وانما

اشتقاق البسط و المقام بشكل منفصل عن بعضهم البعض.



## Text Book Questions



Evaluate the limit if it exist

$$\lim_{t \rightarrow 0} \frac{8^t - 5^t}{t}$$

$$\frac{1-1}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{t \rightarrow 0} \frac{8^t (1) \ln 8 - 5^t (1) \ln 5}{1} = \frac{(1) \ln 8 - (1) \ln 5}{1}$$

$$\therefore \lim_{t \rightarrow 0} \frac{8^t - 5^t}{t} = \ln 8 - \ln 5$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}$$

$$\frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{\sec^2 3x \quad (3)}{\cos 2x \quad (2)} = \frac{(1)^2 (3)}{(1) (2)} = \frac{3}{2}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} = \frac{3}{2}$$



## Text Book Questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$$

$$\frac{1-1-0}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \frac{1-1}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \frac{1}{2}$$



انتبه



يمكن تطبيق اللوبيتال  
أكثر من مرة في نفس  
المسألة

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x}$$

$$\frac{0-0}{0-0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \sec^2 x} = \frac{1-1}{1-1} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{\sin x}{-2 \sec x \sec x \tan x}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{-2 \sec^2 x} = \frac{\sin x}{-2 \frac{1}{\cos^2 x}} = \frac{\sin x \cos^2 x}{-2}$$

$$\lim_{x \rightarrow 0} \frac{\cos x}{-2 \sec^2 x} = \frac{1}{-2} = -\frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x} = -\frac{1}{2}$$

Find

Nov 2016

$$\lim_{x \rightarrow 0^+} \frac{\arcsin \sqrt{x}}{1 - \ln(e - x)}$$

$$\lim_{x \rightarrow 0^+} \frac{\sin^{-1} \sqrt{x}}{1 - \ln(e - x)} \quad \frac{0}{1 - \ln(e)} \quad \frac{0}{1 - 1} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0^+} \frac{\frac{1}{2\sqrt{x}}}{\frac{-1}{e - x}}$$

$$\lim_{x \rightarrow 0^+} \frac{2\sqrt{x} \sqrt{1 - x}}{e - x}$$

$$\lim_{x \rightarrow 0^+} \frac{e - x}{2\sqrt{x} \sqrt{1 - x}} = \frac{e}{0} = \infty$$

$$\therefore \lim_{x \rightarrow 0^+} \frac{\sin^{-1} \sqrt{x}}{1 - \ln(e - x)} = \infty$$



## Text Book Questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} x^3 e^{-x^2}$$

$$\lim_{x \rightarrow \infty} \frac{x^3}{e^{x^2}} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{3x^2}{2xe^{x^2}} = \lim_{x \rightarrow \infty} \frac{3x}{2e^{x^2}} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{3}{2(2x)e^{x^2}} = \frac{3}{\infty} = 0$$

$$\therefore \lim_{x \rightarrow \infty} x^3 e^{-x^2} = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \sin 5x \csc 3x$$

$$\lim_{x \rightarrow 0} \sin 5x \frac{1}{\sin 3x}$$

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{5 \cos 5x}{3 \cos 3x} = \frac{5(1)}{3(1)} = \frac{5}{3}$$

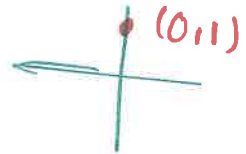
$$\therefore \lim_{x \rightarrow 0} \sin 5x \csc 3x = \frac{5}{3}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (2x - \pi) \sec x$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (2x - \pi) \frac{1}{\cos x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \frac{2x - \pi}{\cos x} = \frac{2 \cdot \frac{\pi}{2} - \pi}{\cos \frac{\pi}{2}} = \frac{0}{0}$$



L'Hop

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \frac{2}{-\sin x} = \frac{2}{-1} = -2$$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}^-} (2x - \pi) \sec x = -2$$

$$\lim_{x \rightarrow 1} \frac{\ln(ex) - e^{x-1}}{(x-1)^2}$$

Past exam

$$\lim_{x \rightarrow 1} \frac{\ln(ex) - e^{x-1}}{(x-1)^2} = \frac{\ln(e(1)) - e^0}{0} = \frac{1-1}{0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\frac{e}{ex} - e^{x-1}}{2(x-1)} = \frac{1-1}{0} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 1} \frac{-\frac{1}{x^2} - e^{x-1}}{2(1)} = \frac{-1-1}{2} = \frac{-2}{2}$$

$$\therefore \lim_{x \rightarrow 1} \frac{\ln(ex) - e^{x-1}}{(x-1)^2} = -1$$

# Evaluate the limit if it exist

past exam

$$\lim_{x \rightarrow 1} \frac{\ln(ex) - 1}{\frac{1}{x} - 1}$$

$$\frac{\ln(e(1)) - 1}{\frac{1}{1} - 1} = \frac{\ln e - 1}{0 - 1} = \frac{1 - 1}{-1} = \frac{0}{-1} = 0$$

Solution #1

L'Hop Rule

$$= \lim_{x \rightarrow 1} \frac{e}{\frac{1}{x} - 1}$$

$$= \lim_{x \rightarrow 1} \frac{e}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{e}{-x^2}$$

$$= \lim_{x \rightarrow 1} \frac{e}{-1} = \frac{e}{-1} = -e$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} \frac{\ln(ex) - 1}{\frac{1}{x} - 1} = -e$$

Solution #2

يمكنك الاستفادة من هوام  $\ln$  قبل الحل

$$= \lim_{x \rightarrow 1} \frac{\ln e + \ln x - 1}{\frac{1}{x} - 1}$$

$$= \lim_{x \rightarrow 1} \frac{1 + \ln x - 1}{\frac{1}{x} - 1} = \frac{1 + 0 - 1}{\frac{1}{1} - 1} = \frac{0}{0}$$

L'Hop. Rule

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 1} \frac{x^2}{-x}$$

$$= \lim_{x \rightarrow 1} \frac{x}{-1} = \frac{1}{-1} = -1$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} \frac{\ln(ex) - 1}{\frac{1}{x} - 1} = -1$$

\*Note\*

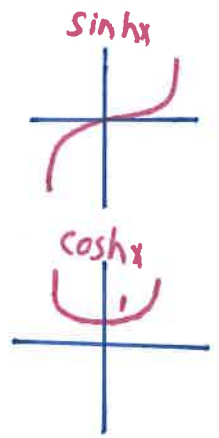
~~$\ln(ex) = x$~~  خطأ

$\ln e^x = x$  ✓ OK

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} \frac{x \sinh x}{1 - \cosh x}$$

$$\frac{0}{1-1} = \frac{0}{0}$$



L'Hospital Rule

$$= \lim_{x \rightarrow 0^+} \frac{\text{اشتقا بشتا} + \text{اشتقا} \quad (1) \sinh x + x \cosh x}{- \sinh x} \quad \frac{0+0}{0} = \frac{0}{0}$$

L'Hospital Rule

$$= \lim_{x \rightarrow 0^+} \frac{\cosh x + (1) \cosh x + x \sinh x}{- \cosh x}$$

$$= \frac{1 + 1 + 0}{-1} = \frac{2}{-1} = -2$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} \frac{x \sinh x}{1 - \cosh x} = -2$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \left( \frac{e^{-x^2} - 1}{x \sin x} \right)$$

$$\frac{e^0 - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0}$$

L, Hop. Rule

$$= \lim_{x \rightarrow 0} \frac{e^{-x^2} (-2x)}{(1) \sin x + x \cos x}$$

اشتق بـ  $x$  + مشتق  $\sin x$

$$\frac{0}{0 + 0} = \frac{0}{0}$$

L, Hop. Rule

$$= \lim_{x \rightarrow 0} \frac{e^{-x^2} (-2x) (-2x) + e^{-x^2} (-2)}{\cos x + (1) \cos x + x (-\sin x)}$$

اشتق بـ  $x$  + مشتق  $x$  + مشتق  $e^{-x^2}$

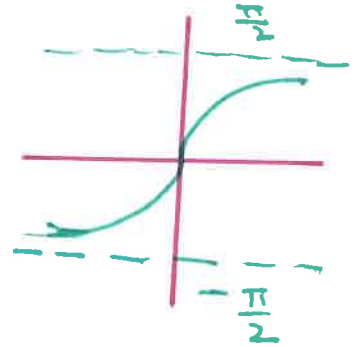
$$= \frac{0 + e^0 (-2)}{1 + 1 + 0} = \frac{-2}{2} = -1$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0} \left( \frac{e^{-x^2} - 1}{x \sin x} \right) = -1$$

past exam

$$\lim_{x \rightarrow \infty} \frac{\tan^{-1} x}{1 + x^2}$$

$$= \frac{\tan^{-1} \infty}{\infty} = \frac{\frac{\pi}{2}}{\infty} = 0$$



\* نصيحة !! جرب !! \*

$$\lim_{x \rightarrow \infty} \frac{\tan^{-1} x}{1 + x^2}$$

$$\frac{\infty}{\infty}$$

التقريب  
هذا

L'Hop Rule

$$= \lim_{x \rightarrow \infty} \frac{1}{x^2 + 1}$$

$$\frac{1}{x^2 + 1}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{2x(x^2 + 1)}$$

$$= \frac{1}{\infty} = 0$$



## Multiple choice



$$\lim_{x \rightarrow 0} \frac{x 2^x}{2^x - 1}$$

Oct 2017

(a) does not exist      (b) equals to  $-\ln 2$

(c) equals to  $\frac{1}{\ln 2}$       (d) equals to  $2\ln 2$

(e) equals to  $\ln 2$

$$\lim_{x \rightarrow 0} \frac{x 2^x}{2^x - 1} \quad \frac{0}{2^0 - 1} = \frac{0}{1 - 1} = \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{(1) 2^x + x 2^x (1) \ln 2}{2^x (1) \ln 2}$$

$$= \frac{1 + 0}{(1) \ln 2} = \frac{1}{\ln 2}$$

(c) ✓

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} \left( \frac{e^x}{\sin x} - \frac{1}{x} \right)$$

$$\frac{1}{0} - \frac{1}{0}$$

$$\infty - \infty$$

$$= \lim_{x \rightarrow 0^+} \left( \frac{x e^x}{x \sin x} - \frac{\sin x}{x \sin x} \right)$$

$$= \lim_{x \rightarrow 0^+} \left( \frac{x e^x - \sin x}{x \sin x} \right) \quad \frac{0-0}{0} \quad \frac{0}{0}$$

L'Hop Rule

$$= \lim_{x \rightarrow 0^+}$$

$$\frac{\begin{matrix} \text{اشتقاق} + \text{بشأ} \\ (1) e^x + x e^x - \cos x \end{matrix}}{\begin{matrix} (1) \sin x + x \cos x \\ \text{اشتقاق} + \text{بشأ} \end{matrix}}$$

$$\frac{1+0-1}{0+0} \quad \frac{0}{0}$$

L'Hop Rule

$$= \lim_{x \rightarrow 0^+}$$

$$\frac{\begin{matrix} \text{اشتقاق} + \text{بشأ} + \text{بشأ} \\ e^x + (1) e^x + x e^x + \sin x \end{matrix}}{\begin{matrix} \cos x + (1) \cos x + x(-\sin x) \\ \text{اشتقاق} + \text{بشأ} + \text{بشأ} \end{matrix}}$$

$$= \frac{1+1+0+0}{1+1+0} = \frac{2}{2} = 1$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} \left( \frac{e^x}{\sin x} - \frac{1}{x} \right) = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \csc x \right)$$

أصلها لمقلوبها حتى  
تسهل عملية التعويض



$$= \lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \frac{1}{\sin x} \right) \quad \frac{\frac{1}{0} - \frac{1}{0}}{\infty - \infty}$$

$$= \lim_{x \rightarrow 0^+} \left( \frac{\sin x}{x \sin x} - \frac{x}{x \sin x} \right)$$

$$= \lim_{x \rightarrow 0^+} \left( \frac{\sin x - x}{x \sin x} \right) \quad \frac{0-0}{0} \quad \frac{0}{0}$$

L'Hop. Rule

$$= \lim_{x \rightarrow 0^+} \frac{\cos x - 1}{(1) \sin x + x \cos x} \quad \frac{1-1}{0+0} \quad \frac{0}{0}$$

الاشتقا + مشتق الاشتقا

L'Hop Rule

$$= \lim_{x \rightarrow 0^+} \frac{-\sin x}{\cos x + (1) \cos x + x(-\sin x)} = \frac{0}{1+1+0} = \frac{0}{2} = 0$$

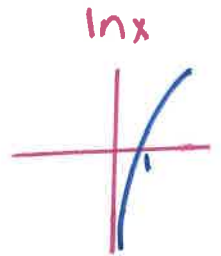
الاشتقا + مشتق الاشتقا

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} \left( \frac{1}{x} - \csc x \right) = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} x^2 \ln x$$

$$0 \ln 0 \\ 0(-\infty)$$



$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x^2}} \quad \frac{\ln 0}{\frac{1}{0}} = \frac{-\infty}{\infty}$$

تم ضمان شكل اللوبيتال  
السبب  
دون الحاجة  
للتعويض



$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-2}}$$

L'Hop. Rule

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-2x^{-3}} \quad \text{مليقاً}$$

$$= \lim_{x \rightarrow 0^+} \frac{x^3}{-2x}$$

$$= \lim_{x \rightarrow 0^+} \frac{x^2}{-2} = \frac{0}{-2} = 0$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \cos x \ln(\cos x)$$

Past exam

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \cos x \ln(\cos x) \quad \cos \frac{\pi}{2} \ln(\cos \frac{\pi}{2})$$

$$0 \ln(0)$$

$$0 (-\infty)$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{\ln(\cos x)}{\frac{1}{\cos x}} \quad \frac{\ln(0)}{\frac{1}{0}} \quad \frac{-\infty}{\infty}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+}$$

$$\frac{\ln(\cos x)}{\sec x}$$

L'Hôpital

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{\frac{-\sin x}{\cos x}}{\sec x \tan x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{-\tan x}{\sec x \tan x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{-1}{\cos x}$$

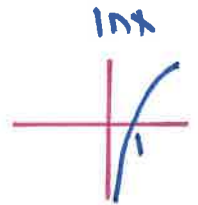
$$\lim_{x \rightarrow \frac{\pi}{2}^+} -\cos x = 0$$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}^+} \cos x \ln(\cos x) = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} \sin x \ln x$$

$$0 \ln 0 \\ 0(-\infty)$$



$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{\sin x}} \quad \frac{\ln 0}{\frac{1}{0}} = \frac{-\infty}{\infty}$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\csc x}$$

تم ضمان شكل اللوبيتال  
البسيط  
ثم البرهان دون الحاجة  
للتعويض



L'Hop Rule

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\csc x \cot x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{\sin x} \frac{1}{\tan x}} \quad \text{فليق}$$

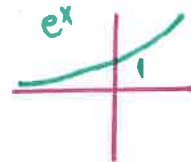
$$= \lim_{x \rightarrow 0^+} \frac{\sin x \tan x}{-x} \quad \frac{0}{0} \quad \text{هلو}$$

L'Hop Rule

$$= \lim_{x \rightarrow 0^+} \frac{\text{اشتقا بـ} + \text{اشتقا بـ}}{\cos x \tan x + \sin x \sec^2 x} = \frac{0+0}{-1} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} \sin x \ln x = 0$$

Evaluate the limit if it exist



$$\lim_{x \rightarrow -\infty} x^2 e^x$$

$$\infty e^{-\infty}$$

$$\infty \cdot 0$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2}{\frac{1}{e^x}} \quad \frac{\infty}{\frac{1}{0}} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2}{e^{-x}}$$

L'Hop Rule

$$= \lim_{x \rightarrow -\infty} \frac{2x}{-e^{-x}} \quad \frac{-\infty}{-\infty} \quad \frac{-\infty}{-\infty}$$

L'Hop Rule

$$= \lim_{x \rightarrow -\infty} \frac{2}{e^{-x}} = \frac{2}{e^{\infty}} = \frac{2}{\infty} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow -\infty} x^2 e^x = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} (\sec x + x)^{\frac{1}{x}}$$

$$(1+0)^{\frac{1}{0}} \quad 1^{\infty}$$

ماله داعي أبطلها  
لمقلوبها لأن الناتج  
معروف



①  $y = (\sec x + x)^{\frac{1}{x}}$

②  $\ln y = \ln (\sec x + x)^{\frac{1}{x}}$

$$\ln y = \frac{1}{x} \ln (\sec x + x)$$

③  $\lim_{x \rightarrow 0^+} \frac{1}{x} \ln (\sec x + x)$

$$= \lim_{x \rightarrow 0^+} \frac{\ln (\sec x + x)}{x} = \frac{\ln (1+0)}{0} = \frac{0}{0}$$

L'Hop Rule

$$= \lim_{x \rightarrow 0^+} \frac{\sec x \tan x + 1}{\sec x + x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sec x \tan x + 1}{\sec x + x} = \frac{0+1}{1+0} = \frac{1}{1} = 1$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} (\sec x + x)^{\frac{1}{x}} = e^1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} (1 + \ln x)^{\frac{1}{x-1}}$$

$$(1+0)^{\frac{1}{0}} \quad 1^{\infty}$$

$$y = (1 + \ln x)^{\frac{1}{x-1}}$$

$$\ln y = \ln (1 + \ln x)^{\frac{1}{x-1}}$$

$$\ln y = \frac{1}{x-1} \ln (1 + \ln x)$$

عند امكانية كتابة  
السؤال على شكل بسط  
ومقام فانه يتم ذلك



$$\lim_{x \rightarrow 1} \frac{1}{x-1} \ln (1 + \ln x)$$

$$= \lim_{x \rightarrow 1} \frac{\ln (1 + \ln x)}{x-1} \quad \frac{\ln(1+0)}{0} \quad \frac{0}{0}$$

L'Hop Rule

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{1+\ln x} \cdot \frac{1}{x}}{1}$$

$$= \lim_{x \rightarrow 1} \frac{1}{(1 + \ln x) x} = \frac{1}{(1+0)(1)} = \frac{1}{1} = 1$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} (1 + \ln x)^{\frac{1}{x-1}} = e^1$$

Find

Fall 2021

$$\lim_{x \rightarrow 0^+} (3e^x - 2)^{\csc x}$$

$$\lim_{x \rightarrow 0^+} (3e^x - 2)^{\frac{1}{\sin x}} \quad (3-2)^{\frac{1}{0}} \quad 1^{\infty}$$

$$y = (3e^x - 2)^{\frac{1}{\sin x}}$$

$$\ln y = \ln (3e^x - 2)^{\frac{1}{\sin x}}$$

$$\ln y = \frac{1}{\sin x} \ln (3e^x - 2)$$

$$\lim_{x \rightarrow 0^+} \frac{\ln (3e^x - 2)}{\sin x} \quad \frac{\ln (3-2)}{0} \quad \frac{\ln 1}{0} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0^+}$$

$$\frac{3e^x}{3e^x - 2} \quad \frac{\cos x}{\cos x (3e^x - 2)}$$

$$\lim_{x \rightarrow 0^+}$$

$$\frac{3e^x}{\cos x (3e^x - 2)} = \frac{3(1)}{(1)(3-2)} = \frac{3}{1} = 3$$

$$\therefore \lim_{x \rightarrow 0^+} (3e^x - 2)^{\csc x} = e^3$$



## Text Book Question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} (4x + 1)^{\cot x}$$

أبطلها لمقلوبها حتى  
تسهل عملية التعويض



$$\lim_{x \rightarrow 0^+} (4x + 1)^{\frac{1}{\tan x}} \quad (0+1)^{\frac{1}{0}} \quad 1^\infty$$

$$y = (4x + 1)^{\frac{1}{\tan x}}$$

$$\ln y = \ln (4x + 1)^{\frac{1}{\tan x}}$$

$$\ln y = \frac{1}{\tan x} \ln (4x + 1)$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(4x + 1)}{\tan x} = \frac{\ln(0+1)}{0} = \frac{0}{0}$$

L'Hop. Rule

$$= \lim_{x \rightarrow 0^+} \frac{4}{\sec^2 x (4x + 1)}$$

$$= \lim_{x \rightarrow 0^+} \frac{4}{\sec^2 x (4x + 1)} = \frac{4}{(1)^2 (0+1)} = \frac{4}{1} = 4$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} (4x + 1)^{\cot x} = e^4$$

AL OTAYBI'S  
LIBRARY

Evaluate the limit

$$\lim_{x \rightarrow 0^+} (\cosh x)^{\cot x}$$

$$\lim_{x \rightarrow 0^+} (\cosh x)^{\frac{1}{\tan x}} \quad \left| \frac{1}{0} \right| \infty$$

$$y = (\cosh x)^{\frac{1}{\tan x}}$$

$$\ln y = \ln (\cosh x)^{\frac{1}{\tan x}}$$

$$\ln y = \frac{1}{\tan x} \ln (\cosh x)$$

$$\lim_{x \rightarrow 0^+} \frac{1}{\tan x} \ln (\cosh x)$$

$$\lim_{x \rightarrow 0^+} \frac{\ln (\cosh x)}{\tan x} \quad \frac{\ln 1}{0} \quad \frac{0}{0}$$

L'HOP.

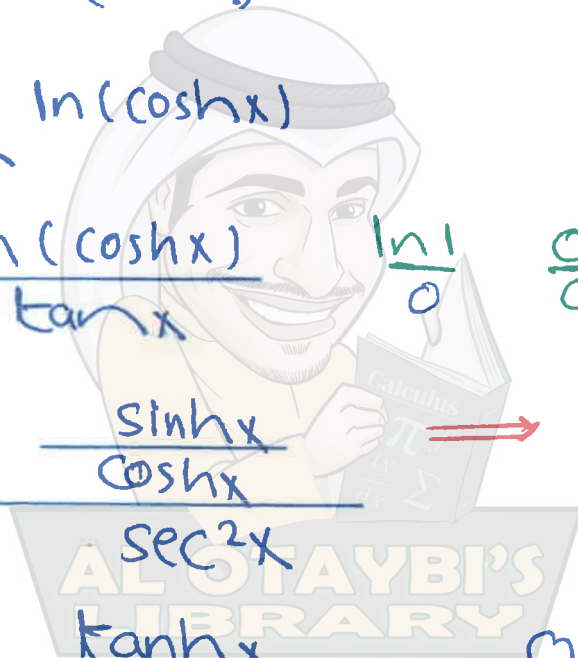
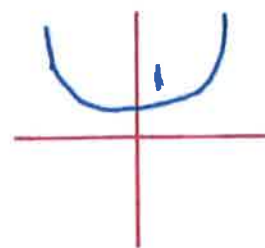
$$\lim_{x \rightarrow 0^+}$$

$$\frac{\sinh x}{\cosh x \sec^2 x}$$

$$\lim_{x \rightarrow 0^+} \frac{\tanh x}{\sec^2 x} = \frac{0}{(1)^2} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} (\cosh x)^{\cot x} = e^0 = 1$$

$\cosh x$



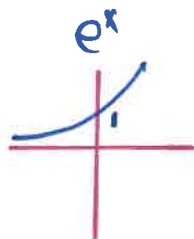
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$$\frac{\sin}{\cos} \rightarrow \tan \quad \text{التانجنت}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} (e^x) e^{-x}$$

$\infty^0$



$$y = (e^x) e^{-x}$$

$$\ln y = \ln (e^x) e^{-x} \quad \begin{cases} \nearrow \ln e^x e^{-x} = x e^{-x} \\ \text{OR} \\ \nwarrow \ln (e^x)^{e^{-x}} = e^{-x} \ln e^x = e^{-x} x \end{cases}$$

$$\lim_{x \rightarrow \infty} x e^{-x} \quad \begin{matrix} \infty e^{-\infty} \\ \infty \cdot 0 \end{matrix}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{\frac{1}{e^{-x}}} \quad \begin{matrix} \frac{\infty}{\frac{1}{0}} & \frac{\infty}{\infty} \end{matrix}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{e^x}$$

L'Hop Rule

$$= \lim_{x \rightarrow \infty} \frac{1}{e^x} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} (e^x) e^{-x} = e^0 = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1^+} \left( \frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{\ln x}}$$

$$\left( \frac{4}{\pi} \tan^{-1} 1 \right)^{\frac{1}{0}} = \left( \frac{4}{\pi} \cdot \frac{\pi}{4} \right)^{\infty} = 1^{\infty}$$

$$y = \left( \frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{\ln x}}$$

$$\ln y = \ln \left( \frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{\ln x}}$$

$$\ln y = \frac{1}{\ln x} \ln \left( \frac{4}{\pi} \tan^{-1} x \right)$$

$$\lim_{x \rightarrow 1^+} \frac{1}{\ln x} \ln \left( \frac{4}{\pi} \tan^{-1} x \right)$$

$$= \lim_{x \rightarrow 1^+} \frac{\ln \left( \frac{4}{\pi} \tan^{-1} x \right)}{\ln x}$$

$\frac{\ln \left( \frac{4}{\pi} \cdot \frac{\pi}{4} \right)}{\ln 1} = \frac{\ln 1}{\ln 1} = \frac{0}{0}$



ممنوع الاختصار بعد التعويض في الحدود الصفرية



L'Hop Rule

$$= \lim_{x \rightarrow 1^+} \frac{\frac{4}{\pi} \tan^{-1} x}{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 1^+} \frac{x}{\tan^{-1} x (x^2 + 1)} = \frac{1}{\frac{\pi}{4} (2)} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1^+} \left( \frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{\ln x}} = e^{\frac{2}{\pi}}$$



## Text Book Question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} (2-x)^{\tan\left(\frac{\pi x}{2}\right)}$$

$$1^{\frac{1}{0}} \quad 1^{\infty}$$

$$y = (2-x)^{\tan\left(\frac{\pi x}{2}\right)}$$

$$\ln y = \ln (2-x)^{\tan\left(\frac{\pi x}{2}\right)}$$

$$\ln y = \tan\left(\frac{\pi x}{2}\right) \ln (2-x)$$

$$\lim_{x \rightarrow 1} \tan\left(\frac{\pi x}{2}\right) \ln(2-x) \quad \infty \cdot 0$$

$$\lim_{x \rightarrow 1} \frac{\ln(2-x)}{\frac{1}{\tan\left(\frac{\pi x}{2}\right)}} \quad \frac{0}{\frac{1}{\infty}} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\ln(2-x)}{\cot\left(\frac{\pi x}{2}\right)}$$

L'Hop

$$\lim_{x \rightarrow 1} \frac{-1}{2-x} \cdot \frac{1}{-\csc^2\left(\frac{\pi x}{2}\right) \cdot \frac{\pi}{2}}$$

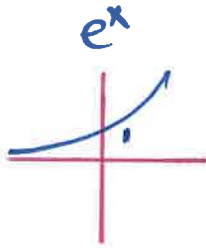
$$\lim_{x \rightarrow 1} \frac{2}{\csc^2\left(\frac{\pi x}{2}\right) \pi (2-x)} = \frac{2}{(1) \pi (1)} = \frac{2}{\pi}$$

$$\therefore \lim_{x \rightarrow 1} (2-x)^{\tan\left(\frac{\pi x}{2}\right)} = e^{\frac{2}{\pi}}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} (1 + e^x)^{\frac{1}{x}}$$

$$(1 + \infty)^{\frac{1}{\infty}} \infty^0$$



$$y = (1 + e^x)^{\frac{1}{x}}$$

$$\ln y = \ln (1 + e^x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (1 + e^x)$$

$$\lim_{x \rightarrow \infty} \frac{\ln (1 + e^x)}{x} \quad \frac{\ln \infty}{\infty} \quad \frac{\infty}{\infty}$$

عند امكانية كتابة السؤال  
على شكل بسط ومقام  
فانه يتم ذلك



L'Hop. Rule

$$= \lim_{x \rightarrow \infty} \frac{e^x}{1 + e^x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{1 + e^x}$$

L'Hop. Rule

$$= \lim_{x \rightarrow \infty} \frac{e^x}{e^x}$$

$$= \lim_{x \rightarrow \infty} 1 = 1$$

أقدر اختصر قبل  
التعويض



$$\Rightarrow \therefore \lim_{x \rightarrow \infty} (1 + e^x)^{\frac{1}{x}} = e^1 = e$$

Evaluate

Fall 2021

$$\lim_{x \rightarrow +\infty} (3e^x + 2x^2)^{4/x}$$

$$\lim_{x \rightarrow \infty} (3e^x + 2x^2)^{\frac{4}{x}} \quad (\infty + \infty)^{\frac{4}{\infty}} \quad \infty^0$$

$$y = (3e^x + 2x^2)^{\frac{4}{x}}$$

$$\ln y = \ln (3e^x + 2x^2)^{\frac{4}{x}}$$

$$\ln y = \frac{4}{x} \ln (3e^x + 2x^2)$$

$$\lim_{x \rightarrow \infty} \frac{4}{x} \ln (3e^x + 2x^2)$$

$$\lim_{x \rightarrow \infty} \frac{4 \ln (3e^x + 2x^2)}{x} \quad \frac{4 \ln (\infty + \infty)}{\infty} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4 \frac{3e^x + 4x}{3e^x + 2x^2}}{1} = \lim_{x \rightarrow \infty} \frac{4(3e^x + 4x)}{3e^x + 2x^2} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4(3e^x + 4)}{3e^x + 4x} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{4(3e^x)}{3e^x + 4} = \lim_{x \rightarrow \infty} \frac{12e^x}{3e^x + 4} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{12e^x}{3e^x} = 4 \Rightarrow \lim_{x \rightarrow \infty} (3e^x + 2x^2)^{\frac{4}{x}} = e^4$$

$$\lim_{x \rightarrow 1^+} (\ln x)^{\ln x}$$

$0^0$

$$y = (\ln x)^{\ln x}$$

$$\ln y = \ln (\ln x)^{\ln x}$$

$$\ln y = \ln x \ln (\ln x)$$

مايفرق منواترل

$$\lim_{x \rightarrow 1^+} \ln x \ln (\ln x)$$

$$\ln 1 \ln (\ln 1)$$

$$0 \ln 0$$

$$0 (-\infty)$$

$$\lim_{x \rightarrow 1^+} \frac{\ln (\ln x)}{\frac{1}{\ln x}}$$

$$\frac{\ln (\ln x)}{\frac{1}{\ln x}}$$

$$\frac{\ln (\ln 1)}{\frac{1}{\ln 1}} = \frac{\ln 0}{\frac{1}{0}} = \frac{0}{\infty} = 0$$

$$\lim_{x \rightarrow 1^+} \frac{\ln (\ln x)}{(\ln x)^{-1}}$$

$$\frac{\ln (\ln x)}{(\ln x)^{-1}}$$

L'HOP

$$\lim_{x \rightarrow 1^+} \frac{\ln (\ln x)}{(\ln x)^{-1}}$$

$$\frac{\frac{1}{\ln x}}{- (\ln x)^{-2}}$$

$$\lim_{x \rightarrow 1^+} \frac{(\ln x)^2}{- \ln x}$$

$$\frac{(\ln x)^2}{- \ln x}$$

$$= \frac{\ln 1}{-1} = \frac{0}{-1} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1^+} (\ln x)^{\ln x} = e^0 = 1$$



مهم جدا جدا



$$\lim_{x \rightarrow \infty} \left( \frac{x-1}{x+1} \right)^{3x+2}$$

$$\left( \frac{\infty}{\infty} \right)^{\infty}$$

$$y = \left( \frac{x-1}{x+1} \right)^{3x+2}$$

$$\ln y = \ln \left( \frac{x-1}{x+1} \right)^{3x+2}$$

$$\ln y = (3x+2) \ln \left( \frac{x-1}{x+1} \right)$$

$$\lim_{x \rightarrow \infty} (3x+2) \ln \left( \frac{x-1}{x+1} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{\ln \left( \frac{x-1}{x+1} \right)}{\frac{1}{3x+2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\ln(x-1) - \ln(x+1)}{(3x+2)^{-1}}$$

L'Hop. Rule

$$\lim_{x \rightarrow \infty} \left( \frac{+}{+} \right)^{3x+2} = 1^{\infty}$$

$$\begin{matrix} \infty \ln 1 \\ \infty \cdot 0 \end{matrix}$$

$$\frac{\ln 1}{\frac{1}{\infty}} = \frac{0}{0}$$

L'Hop. Rule

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x-1} - \frac{1}{x+1}}{- (3x+2)^{-2} (3)}$$

مليق

$$\lim_{x \rightarrow \infty} \frac{\frac{x+1}{(x-1)(x+1)} - \frac{x-1}{(x+1)(x-1)}}{-3(3x+2)^{-2}}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{x+1 - [x-1]}{(x-1)(x+1)}}{-3(3x+2)^{-2}}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{\cancel{x+1} - \cancel{x} + 1}{x^2 - 1}}{-3(3x+2)^{-2}} \quad (2)$$

$$\lim_{x \rightarrow \infty} \frac{2(3x+2)^2}{-3(x^2-1)} \quad \frac{\infty}{\infty}$$

انا اللمت (1) تسع ال  $\pm \infty$

(2) حسب درجة الحدودية

\* نفس الدرجة البسط والمقام \*

معاملات أكبر الأسس  $\Rightarrow$

$$\lim_{x \rightarrow \infty} \frac{2[9x^2 + 12x + 4]}{-3(x^2 - 1)}$$

$$= \frac{2(9)}{-3} = 2(-3) = -6$$

$$\Rightarrow \infty \lim_{x \rightarrow \infty} \left( \frac{x-1}{x+1} \right)^{3x+2} = e^{-6}$$



## Text Book Question



Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \left( \frac{2x-3}{2x+5} \right)^{2x+1}$$

$$\left( \frac{\infty}{\infty} \right)^{\infty}$$

$$y = \left( \frac{2x-3}{2x+5} \right)^{2x+1}$$

$$\ln y = \ln \left( \frac{2x-3}{2x+5} \right)^{2x+1}$$

$$\ln y = (2x+1) \ln \left( \frac{2x-3}{2x+5} \right)$$

$$\lim_{x \rightarrow \infty} (2x+1) \ln \left( \frac{2x-3}{2x+5} \right) \quad \begin{matrix} \infty \ln 1 \\ \infty \cdot 0 \end{matrix}$$

$$\lim_{x \rightarrow \infty} \frac{\ln \left( \frac{2x-3}{2x+5} \right)}{\frac{1}{2x+1}} \quad \frac{\ln 1}{\frac{1}{\infty}} = \frac{0}{0}$$

$$\lim_{x \rightarrow \infty} \frac{\ln(2x-3) - \ln(2x+5)}{(2x+1)^{-1}}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{\frac{2}{2x-3} - \frac{2}{2x+5}}{- (2x+1)^{-2} \cdot 2}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{2(2x+5) - 2(2x-3)}{(2x-3)(2x+5)}}{-2(2x+1)^{-2}}$$

$$\lim_{x \rightarrow \infty} \frac{4x+10 - 4x+6}{-2(2x+1)^{-2}(2x-3)(2x+5)}$$

$$\lim_{x \rightarrow \infty} \frac{16(2x+1)^2}{-2(2x-3)(2x+5)} = \lim_{x \rightarrow \infty} \frac{16(4x^2+4x+1)}{-2(2x-3)(2x+5)}$$

$$= \frac{16(4)}{-2(2)(2)} = -8 \implies \lim_{x \rightarrow \infty} \left( \frac{2x-3}{2x+5} \right)^{2x+1} = e^{-8}$$

L'Hop

$$\lim_{x \rightarrow \infty} \left( \frac{2}{2} \right)^{2x+1}$$

$\infty$



مهم جدا جدا



$$\lim_{x \rightarrow \infty} \left( \frac{x+3}{\cosh x} \right)^{\frac{1}{x}}$$

$$\left( \frac{\infty}{\infty} \right)^{\frac{1}{\infty}}$$



$$y = \left( \frac{x+3}{\cosh x} \right)^{\frac{1}{x}}$$

$$\ln y = \ln \left( \frac{x+3}{\cosh x} \right)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln \left( \frac{x+3}{\cosh x} \right)$$

$$\lim_{x \rightarrow \infty} \frac{\ln \left( \frac{x+3}{\cosh x} \right)}{x} \quad \frac{\ln 0}{\infty} = \frac{-\infty}{\infty}$$

L'Hop. Rule

$$\lim_{x \rightarrow \infty} \left( \frac{1}{\sinh x} \right)^{\frac{1}{x}}$$

$$\left( \frac{1}{\infty} \right)^{\frac{1}{\infty}} = 0^0$$

$\sinh x$

$$= \lim_{x \rightarrow \infty} \frac{\ln(x+3) - \ln \cosh x}{x}$$

L'Hop. Rule

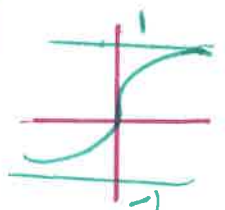
$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x+3} - \frac{\sinh x}{\cosh x}}{1}$$

$$= \lim_{x \rightarrow \infty} \left[ \frac{1}{x+3} - \tanh x \right] = \frac{1}{\infty} - \tanh \infty$$

$$= 0 - 1$$

$$= -1$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \left( \frac{x+3}{\cosh x} \right)^{\frac{1}{x}} = e^{-1}$$



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \left( \frac{\sin 2x}{e^x - 1} \right)^{\tan^{-1} x}$$

$$\left( \frac{0}{0} \right)^0$$

L'Hop

$$\lim_{x \rightarrow 0} \left( \frac{2 \cos 2x}{e^x} \right)^{\tan^{-1} x} = \left( \frac{2(1)}{1} \right)^0 = 1$$

$$\therefore \lim_{x \rightarrow 0} \left( \frac{\sin x}{e^x - 1} \right)^{\tan^{-1} x} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} e^{\left( \frac{1 - \cos x}{\sin x} \right)}$$

$$e^{\frac{1-1}{0}} = e^{\frac{0}{0}}$$

L'Hop

$$\lim_{x \rightarrow 0} e^{\frac{\sin x}{\cos x}} = e^{\frac{0}{1}} = e^0 = 1$$

$$\therefore \lim_{x \rightarrow 0} e^{\left( \frac{1 - \cos x}{\sin x} \right)} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1^+} [\ln(x^7 - 1) - \ln(x^5 - 1)]$$

$$\lim_{x \rightarrow 1^+} \ln \left( \frac{x^7 - 1}{x^5 - 1} \right) \quad \ln \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 1^+} \ln \frac{7x^6}{5x^4} = \ln \frac{7}{5}$$

$$\therefore \lim_{x \rightarrow 1^+} [\ln(x^7 - 1) - \ln(x^5 - 1)] = \ln \frac{7}{5}$$

If  $f$  is differentiable function with  $f(2) = 0$  and  $f'(2) = 7$

Then evaluate

$$\lim_{x \rightarrow 0} \frac{f(2+3x)}{2x}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{f(2+3x)}{2x} \quad \frac{f(2+0)}{0} \quad \frac{f(2)}{0} \quad \frac{0}{0} \\ & \text{L'Hop} \quad \lim_{x \rightarrow 0} \frac{f'(2+3x)(3)}{2} = \frac{f'(2+0)(3)}{2} = \frac{f'(2)(3)}{2} = \frac{7(3)}{2} \\ \Rightarrow \therefore \lim_{x \rightarrow 0} \frac{f(2+3x)}{2x} &= \frac{21}{2} \quad \text{Tricky question} \quad = \frac{21}{2} \end{aligned}$$

Find the value of  $a$  for which

$$\lim_{x \rightarrow 0} \frac{\tan x + ax}{x^2}$$

exist, then find the limit for this value of  $a$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\tan x + ax}{x^2} \quad \frac{0+0}{0} \quad \frac{0}{0} \\ & \text{L'Hop} \quad \lim_{x \rightarrow 0} \frac{\sec^2 x + a}{2x} \quad \frac{1+a}{0} \end{aligned}$$

The limit is exist only when  $1+a=0 \rightarrow \boxed{a=-1}$

$$\text{Now} \quad \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{2x} \quad \frac{1-1}{0} \quad \frac{0}{0}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow 0} \frac{2\sec x \sec x \tan x}{2} = \frac{0}{2} = 0$$

$$\therefore \lim_{x \rightarrow 0} \frac{\tan x + ax}{x^2} = 0 \quad \text{when } a = -1$$

Find  $b$  such that

Past exam

$$\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x + bx^3}{x^3} = 0$$

$$\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x + bx^3}{x^3} \quad \frac{0 - 0 + 0}{0} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{3 \cos(3x) - 3 + 3bx^2}{3x^2} \quad \frac{3(1) - 3 + 0}{0} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{3(-3 \sin(3x)) - 0 + 6bx}{6x}$$

$$\lim_{x \rightarrow 0} \frac{-9 \sin(3x) + 6bx}{6x} \quad \frac{0 + 0}{0} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow 0} \frac{-27 \cos(3x) + 6b}{6} = \boxed{\frac{-27(1) + 6b}{6}}$$

$$\Rightarrow \therefore \frac{-27 + 6b}{6} = 0$$

$\times 6$

$$-27 + 6b = 0$$

$$6b = 27$$

$$b = \frac{27}{6} \rightarrow \therefore$$

$$\boxed{b = \frac{9}{2}}$$

Find the value of  $a$  so that

past exam

$$\lim_{x \rightarrow \infty} (\sinh(ax))^{\frac{1}{x}} = 3$$

$$\lim_{x \rightarrow \infty} (\sinh(ax))^{\frac{1}{x}} \quad (\sinh \infty)^{\frac{1}{\infty}} \quad \infty^0$$

$$y = (\sinh(ax))^{\frac{1}{x}}$$

$$\ln y = \ln (\sinh(ax))^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (\sinh(ax))$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} \ln (\sinh(ax))$$

$$\lim_{x \rightarrow \infty} \frac{\ln (\sinh(ax))}{x} \quad \frac{\ln \infty}{\infty} \quad \frac{\infty}{\infty}$$

L'Hop.

$$\lim_{x \rightarrow \infty}$$

$$\frac{\cosh(ax)}{\sinh(ax)}$$

$$\coth(ax) \quad a$$

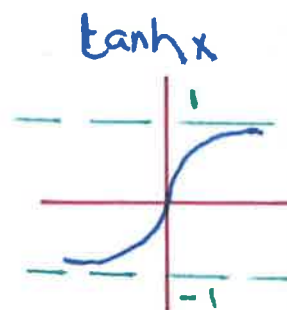
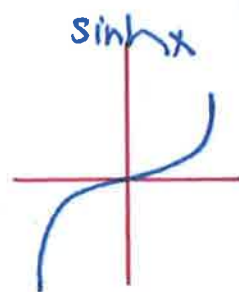
$$\lim_{x \rightarrow \infty}$$

$$\frac{1}{\tanh(ax)} \quad a$$

$$\lim_{x \rightarrow \infty} \frac{a}{\tanh(ax)} = \frac{a}{\tanh \infty} = \frac{a}{1} = a$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} (\sinh(ax))^{\frac{1}{x}} = e^a$$

$$\Rightarrow e^a = 3 \rightarrow \ln e^a = \ln 3 \rightarrow a = \ln 3$$





استثناء

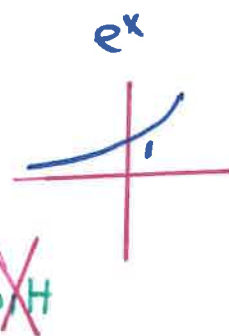
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$$\lim_{x \rightarrow \infty} \frac{3e^x + 2e^{-x}}{2e^x - 3e^{-2x}}$$

$$\frac{3e^\infty + 0}{2e^\infty - 0}$$

g/g



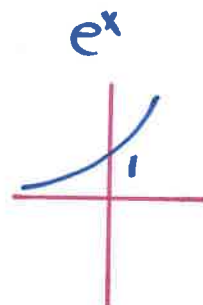
$$= \lim_{x \rightarrow \infty} \frac{\frac{3e^x}{e^x} + \frac{2e^{-x}}{e^x}}{\frac{2e^x}{e^x} - \frac{3e^{-2x}}{e^x}}$$

$$= \lim_{x \rightarrow \infty} \frac{3 + 2e^{-2x}}{2 - 3e^{-3x}} = \frac{3 + 0}{2 - 0} = \frac{3}{2}$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \frac{3e^x + 2e^{-x}}{2e^x - 3e^{-2x}} = \frac{3}{2}$$

$$\lim_{x \rightarrow \infty} \frac{e^{5x} + e^{2x}}{4e^{5x} + 5e^{3x}}$$

$$\frac{\infty + \infty}{\infty + \infty}$$



$$= \lim_{x \rightarrow \infty} \frac{\frac{e^{5x}}{e^{5x}} + \frac{e^{2x}}{e^{5x}}}{4 \frac{e^{5x}}{e^{5x}} + 5 \frac{e^{3x}}{e^{5x}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1 + e^{-3x}}{4 + 5e^{-2x}} = \frac{1 + 0}{4 + 0} = \frac{1}{4}$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \frac{e^{5x} + e^{2x}}{4e^{5x} + 5e^{3x}} = \frac{1}{4}$$



Good to know



$$\lim_{x \rightarrow \infty} a^x \begin{cases} = \infty, & \text{if } a > 1 \\ = 0, & \text{if } |a| < 1 \end{cases}$$

### Examples

$$\lim_{x \rightarrow \infty} 3^x$$

$$= \infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{1}{3}\right)^x$$

$$= 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{3^x - 5^x}{6^x + 4^x}$$

$$\frac{\infty - \infty}{\infty + \infty} \quad \frac{\infty - \infty}{\infty}$$

$\div 6^x$

$$\lim_{x \rightarrow \infty}$$

$$\frac{\frac{3^x}{6^x} - \frac{5^x}{6^x}}{\frac{6^x}{6^x} + \frac{4^x}{6^x}}$$

$$\lim_{x \rightarrow \infty}$$

$$\frac{\left(\frac{3}{6}\right)^x - \left(\frac{5}{6}\right)^x}{1 + \left(\frac{4}{6}\right)^x} = \frac{0 - 0}{1 + 0} = \frac{0}{1} = 0$$

$$\therefore \lim_{x \rightarrow \infty} \frac{3^x - 5^x}{6^x + 4^x} = 0$$

$$\lim_{x \rightarrow \infty} \sqrt[x]{3^x + 4^x}$$

$$\lim_{x \rightarrow \infty} (3^x + 4^x)^{\frac{1}{x}} \quad (\infty + \infty)^{\frac{1}{\infty}} \quad \infty^0$$

$$y = (3^x + 4^x)^{\frac{1}{x}}$$

$$\ln y = \ln (3^x + 4^x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (3^x + 4^x)$$

$$\lim_{x \rightarrow \infty} \frac{\ln (3^x + 4^x)}{x} \quad \frac{\ln (\infty + \infty)}{\infty} \quad \frac{\ln \infty}{\infty} \quad \frac{\infty}{\infty}$$

L.Hop

$$\lim_{x \rightarrow \infty} \frac{3^x \ln 3 + 4^x \ln 4}{3^x + 4^x}$$

$$\lim_{x \rightarrow \infty} \frac{3^x \ln 3 + 4^x \ln 4}{3^x + 4^x}$$

$\div 4^x$

$$\lim_{x \rightarrow \infty} \frac{\frac{3^x}{4^x} \ln 3 + \ln 4}{\frac{3^x}{4^x} + 1}$$

$$\lim_{x \rightarrow \infty} \frac{\left(\frac{3}{4}\right)^x \ln 3 + \ln 4}{\left(\frac{3}{4}\right)^x + 1}$$

$$= \frac{0 + \ln 4}{0 + 1} = \ln 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \sqrt[x]{3^x + 4^x} = e^{\ln 4} = 4$$

قراءة



## Tricky Question

استثناء 1



$$\lim_{x \rightarrow \infty} \frac{\log_2(e^{2x} + 1)}{\cosh x}$$



$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{\ln 2} \ln(e^{2x} + 1)}{\cosh x}$$

$$= \frac{1}{\ln 2} \lim_{x \rightarrow \infty} \frac{\ln(e^{2x} + 1)}{\cosh x} \quad \frac{\ln(\infty)}{\cosh \infty} \quad \frac{\infty}{\infty}$$

L'Hop. Rule

$$= \frac{1}{\ln 2} \lim_{x \rightarrow \infty} \frac{2e^{2x}}{e^{2x} + 1} \cdot \frac{1}{\sinh x}$$

$$= \frac{1}{\ln 2} \lim_{x \rightarrow \infty} \frac{2e^{2x}}{(e^{2x} + 1) \sinh x}$$

$\div e^{2x}$

$$= \frac{1}{\ln 2} \lim_{x \rightarrow \infty} \frac{2 \frac{e^{2x}}{e^{2x}}}{(\frac{e^{2x}}{e^{2x}} + 1) \sinh x}$$

$$= \frac{1}{\ln 2} \lim_{x \rightarrow \infty} \frac{2}{(1 + e^{-2x}) \sinh x} = \frac{1}{\ln 2} \frac{2}{(1+0)\infty} = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{\log_2(e^{2x} + 1)}{\cosh x} = 0$$



استثناء

2



$$\lim_{x \rightarrow \infty} \frac{3x^5 + 3x^4 + 5}{2x^5 + 4x^3 + 10}$$

8/8

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^5} \left( 3 + \frac{3}{x} + \frac{5}{x^5} \right)}{\cancel{x^5} \left( 2 + \frac{4}{x^2} + \frac{10}{x^5} \right)}$$

$$= \frac{3 + 0 + 0}{2 + 0 + 0} = \frac{3}{2}$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \frac{3x^5 + 3x^4 + 5}{2x^5 + 4x^3 + 10} = \frac{3}{2}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 5}}{x + 3}$$

8/8

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{\cancel{x^2} \left( 4 + \frac{5}{x^2} \right)}}{x + 3}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{4 + \frac{5}{x^2}}}{x + 3}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{4 + \frac{5}{x^2}}}{\cancel{x} \left[ 1 + \frac{3}{x} \right]} = \frac{\sqrt{4+0}}{1+0} = \frac{2}{1} = 2$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 5}}{x + 3} = 2$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0^+} x e^{\frac{1}{x}}$$

$$\begin{matrix} 0e^{\frac{1}{0}} \\ 0e^{\infty} \\ 0\infty \end{matrix}$$

$$\lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x}}}{\frac{1}{x}} \quad \frac{e^{\frac{1}{0}}}{\frac{1}{0}} \quad \frac{e^{\infty}}{\frac{1}{0}} \quad \frac{\infty}{0}$$

L'Hop

$$\lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x}} (\cancel{\frac{1}{x^2}})}{\cancel{\frac{1}{x^2}}} = e^{\frac{1}{0}} = e^{\infty} = \infty$$

$$\Rightarrow \therefore \lim_{x \rightarrow 0^+} x e^{\frac{1}{x}} = \infty$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} (x - x e^{\frac{1}{x}})$$

$$\begin{matrix} \infty - \infty e^{\frac{1}{0}} \\ \infty - \infty e^0 \\ \infty - \infty \end{matrix}$$

$$\lim_{x \rightarrow \infty} x(1 - e^{\frac{1}{x}}) \quad \begin{matrix} \infty(1 - e^{\frac{1}{0}}) \\ \infty(1 - e^0) \\ \infty(1 - 1) \\ \infty \cdot 0 \end{matrix}$$

$$\lim_{x \rightarrow \infty} \frac{1 - e^{\frac{1}{x}}}{\frac{1}{x}} \quad \frac{1 - e^0}{\frac{1}{0}} \quad \frac{0}{0}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{-e^{\frac{1}{x}} (\cancel{\frac{1}{x^2}})}{\cancel{\frac{1}{x^2}}} = -e^{\frac{1}{0}} = -e^0 = -1$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} (x - x e^{\frac{1}{x}}) = -1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow \infty} \frac{1 + \cos x}{x}$$

$$\lim_{x \rightarrow \infty} \left( \frac{1}{x} + \frac{\cos x}{x} \right)$$

إزالة  
 $\cos \infty \sin \infty$   
 $\rightarrow$  Sandwich  $\leftarrow$

$$\lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{\cos x}{x}$$

$$\frac{\cos \infty}{\infty}$$

$$= \frac{1}{\infty}$$

$$= 0$$

$$-1 \leq \cos x \leq 1$$

$$\frac{-1}{x} \leq \frac{\cos x}{x} \leq \frac{1}{x}$$

$$\lim_{x \rightarrow \infty} \frac{-1}{x} = \frac{-1}{\infty} = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = \frac{1}{\infty} = 0$$

$\therefore$  By Sandwich Thm

$$\lim_{x \rightarrow \infty} \frac{\cos x}{x} = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow \infty} \frac{1 + \cos x}{x} = 0 + 0 = 0$$



## Quiz #1



$$\int (\sin 5x + \tan 2x) dx$$

$$= \frac{-\cos 5x}{5} + \frac{\ln |\sec 2x|}{2} + C$$

the same  
 $-\frac{\ln |\cos 2x|}{2}$

$$\int (2 + 2x + e^{2x} - 2^{7x}) dx$$

$$= 2x + \frac{2x^2}{2} + \frac{e^{2x}}{2} - \frac{2^{7x}}{7 \ln 2} + C$$

$$\int \left( \frac{\cos x}{\sin x + 3} + \frac{3 \cos x}{\sin x + 3} - \frac{\cos x}{3 \sin x + 3} \right) dx$$

$$\int \left( \frac{\cos x}{\sin x + 3} + \frac{3 \cos x}{\sin x + 3} - \frac{\frac{1}{3} \cos x}{3 \sin x + 3} \right) dx$$

$$= \ln |\sin x + 3| + 3 \ln |\sin x + 3| - \frac{1}{3} \ln |3 \sin x + 3| + C$$

$$\int \frac{e^{2x}}{e^{2x} + 9} dx$$

$$\frac{1}{2} \int \frac{2e^{2x}}{e^{2x} + 9} dx = \frac{1}{2} \ln |e^{2x} + 9| + C$$

$$\int \frac{e^x}{e^{2x} + 9} dx = \int \frac{e^x}{(e^x)^2 + 9} dx$$

Let  $u = e^x$   
 $du = e^x dx$

$$\int \frac{1}{u^2 + 9} du = \frac{1}{3} \tan^{-1} \left( \frac{u}{3} \right) + C$$

$$= \frac{1}{3} \tan^{-1} \left( \frac{e^x}{3} \right) + C$$



## Quiz #1



$$\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} dx$$

Let  $u = \sin^{-1}x$   
 $du = \frac{1}{\sqrt{1-x^2}} dx$

$$\int e^u du$$

$$= e^u + C$$

$$= e^{\sin^{-1}x} + C$$

$$\int e^x \sec(e^x) dx$$

Let  $u = e^x$   
 $du = e^x dx$

$$\int \sec u du$$

$$= \ln |\sec u + \tan u| + C$$

$$= \ln |\sec e^x + \tan e^x| + C$$

Find  $\frac{dy}{dx}$

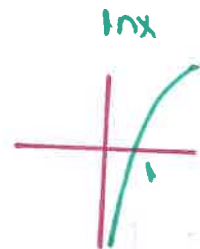
$$y = \cos^{-1}(\ln x) + \sec^{-1}e^x$$

$$y' = \frac{-\frac{1}{x}}{\sqrt{1-(\ln x)^2}} + \frac{e^x}{e^x \sqrt{(e^x)^2 - 1}}$$

Evaluate the limit

$$\lim_{x \rightarrow \infty} \ln x$$

$$= \ln \infty = \infty$$



$$\lim_{x \rightarrow \infty} \tan^{-1} x$$

$$= \tan^{-1} \infty = \frac{\pi}{2}$$

